Tutorial 2 – Fundamental knowledge

T2-1. Calculate the concentration in $[mg/L]$ of a solution with a density of $\rho = n\ g/mL$ and a concentration of 1 ppm (diluted solution).

1 ppm solution with ρ = *n* g/mL,

The concentration is

$$1\ \text{ppm} = \frac{mass\ of\ the\ solute}{mass\ of\ the\ solution} \times 10^6$$

$$1\rho = \frac{mass\ of\ the\ solution}{Volume of\ the\ solution}$$

$\Rightarrow$ ppm × ρ =

$$\frac{mass\ of\ the\ solute}{\cancel{mass\ of\ the\ solution}} \times \frac{\cancel{mass\ of\ the\ solution}}{Volume of\ the\ solution} = \frac{mass\ of\ the\ solute}{Volume of\ the\ solution}$$

So, the concentration of solution with 1 ppm and ρ = *n* g/mL is:

1 ppm × ρ = 1 ppm × *n* g/mL = $10^{-6} \times n\ 10^3$ mg/ 10^{-3} L

= *n* mg/L

When ρ =1 g/mL, 1 ppm is equal to 1 mg/L.

Summary: due to ppm, ppb and % are dimensionless. So, it needs to multiply the density when converting it into the unit with dimension.

In short, 1 ppm · 1 g/mL = 1×10^{-6} g/mL = 1 μg/mL = 1 mg/L

T2-2. How many mL of concentered HCl are needed to make a 500 mL solution with a concentration of 0.250 M HCl? The concentrated HCl is 37.2% (*w*/*w*) and has a density of 1.188 g/cm^3. (MW_{HCl} = 36.46 g/mol)

First, we need to know how much the molarity of the concentrated HCl.

As known, the % composition is 37.2% (w/w) means $\frac{37.2\ g\ of\ HCl}{100\ g\ solution}$

ρ =1.188 g/cm^3 = 1.188 g/mL= $\frac{1.188\ g\ of\ solution}{1\ mL\ of\ solution}$

MW (HCl) = 36.46 g/mol = $\frac{36.46\ g\ HCl}{1\ mol\ HCl}$

⇒ Molarity of HCl ($\frac{mol\ HCl}{\text{Volume of solution}}$) = % composition × ρ ÷ MW (HCl)

$= \frac{37.2\ \cancel{g\ of\ HCl}}{100\ \cancel{g\ solution}} \times \frac{1.188\ \cancel{g\ of\ solution}}{1\ mL\ of\ solution} \div \frac{36.46\ \cancel{g\ HCl}}{1\ mol\ HCl}$

= 0.01212 mol/mL = 12.12 mol/L =12.12 M

The we use the dilution formula (Diluted solution is made of stock solution),

$$M_{stock} \times V_{stock} = M_{dilut} \times V_{dilut}$$

12.12 M ×V_{stock} = 0.250 M × 500 mL

⇒ V_{stock} = 10.3 mL

Summary: Carrying along the units when you do calculations will help you understand some complex questions.

T2-3. A limiting reagent in a chemical reaction is the reagent that is consumed first. Once the limiting reagent is gone, the reaction ceases. To decide which reagent is the limiting one:

a. Find the number of moles of each available reactant.

b. Compare the number of moles present to the number of moles required for a complete reaction.

c. Determine the limiting reagent, which is less moles provided or produce less product.

Determine the limiting reagent in a reaction of 1.00 g $CaCl_2$ (110.98 g/mol) with 1.15 g $Na_2C_2O_4$ (134.00 g/mol) in water. Calculate the unreacted reagent fraction (in %)?

$Na_2C_2O_4$ mol = $\frac{1.15\ g}{134\ g/mol}$ = 8.58 mmol

$CaCl_2$ mol = $\frac{1.00\ g}{110.98\ g/mol}$ = 9.01 mmol

Write down the equation for this reaction:

$$Na_2C_2O_4 + CaCl_2 \rightarrow CaC_2O_4 + 2NaCl$$

Initial (mmol)	8.58	9.01	0	0
Changed (mmol)	8.58	8.58	8.58	2×8.58
Final (mmol)	0	0.43	8.58	17.16

The stoichiometry for $Na_2C_2O_4$ and $CaCl_2$ are 1 and 1, respectively. And the moles of $Na_2C_2O_4$ is less than the moles of $CaCl_2$. So, $Na_2C_2O_4$ is limiting reagent.

The fraction of unreacted $CaCl_2$ = $\frac{0.43}{9.01} \times 100\%$ = 4.77%

Summary: Procedures to find limiting reagent,

Step 1 Calculation moles;

Step 2 Write the equation;

Step 3 Compare with moles;

Step 4 Determine limiting reagent.

T2-4. Check the validity of the statement "In every single breath we take, we inhale at least one molecule from the dying breath of Julius Caesar". Make the following assumptions:

a. Air particles are distributed evenly all over the atmosphere.

b. No particles are created or destroyed.

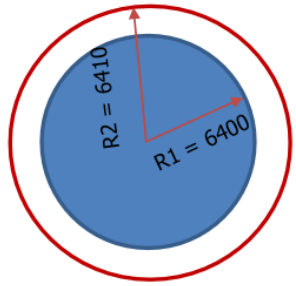
c. The earth radius is 6400 km.

d. The atmosphere height is 10 km.

e. The average volume of air in a single breath is 0.5 L.

f. The air follows the ideal gases law (PV = nRT, R = 0.082 $\frac{\text{L×atm}}{\text{mol×K}}$ = 0.08314 $\frac{\text{L×bar}}{\text{mol×K}}$).

g. The earth is under standard conditions (T = 298 K, P = 1 atm).



Step 1: The volume of the atmosphere is

$V_{atm} = \frac{4}{3}\pi R_2^3 - \frac{4}{3}\pi R_1^3 = \frac{4}{3} \times 3.14 \times [(6410\times10^3)^3 - (6400\times10^3)^3]\ m^3$

$= 5.153 \times 10^{18}\ m^3 = 5.153 \times 10^{21}\ L$

Step 2: How many molecules for the Julius's last breath?

$PV = nRT \Rightarrow n = \frac{PV}{RT} = \frac{1\ \cancel{atm} \times 0.5\ \cancel{L}}{0.082\ \frac{\cancel{L}\times\cancel{atm}}{mol\times\cancel{K}}\times 298\ \cancel{K}} = 0.02\ mol$

$0.02\ mol \times 6.02 \times 10^{23}$ molecules/mol $= 1.204 \times 10^{22}$ molecules

Step 3: How many molecules distributed over the atmosphere?

$= \frac{number\ of\ molecules}{\text{Volume of atmosphere}} = \frac{1.204\times10^{22}\ \text{molecules}}{5.153\times10^{21}\ L} = 2.34$ molecules/L

Step 4: We set *x* molecules of Julius we inhale in one breath from atmosphere, as the particles are not created and disappeared. So, the total molecule from last breath and distributed in the atmosphere should be equated.

$$1.204 \times 10^{22} = \frac{5.153\times10^{21}\ L}{0.5\ L}\,x$$

$$x = 1.17\ \text{molecules}$$

So, in every single breath we take in, we inhale at least one molecule from the dying breath of Julius Caesar is valid.

Do you have another way to demonstrate the validity of the statement?